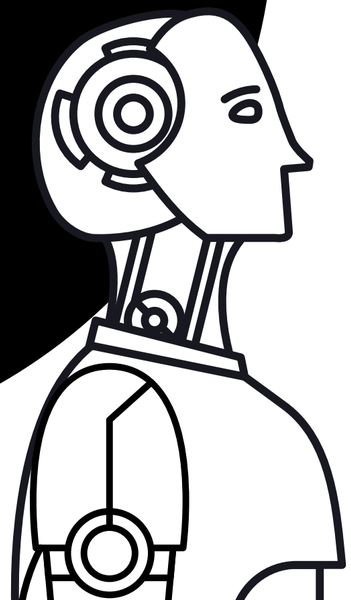
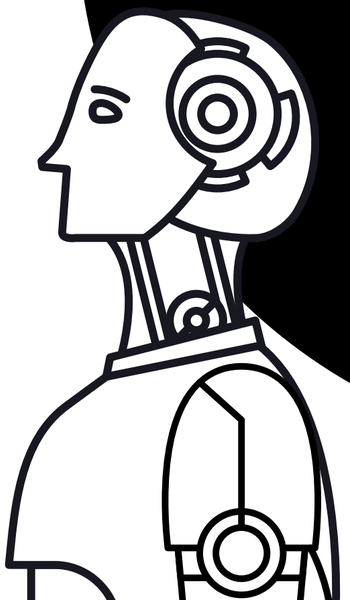
CPE241 FINAL PROJECT



E-COMMERCE WEBSITE

BY GROUP DBM

APPLICATION OF DATABASE DESIGN



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REQUIREMENTS

Summary of requirements and business rules

01

USER MANAGEMENT

- **Users** register with personal information to place **orders**
- **Role-based access system** (Super Admin, Data Entry Admin, Order Manager, Customer)

02

ORDER PROCESSING

- **Each order links to one user**; users can have multiple orders
- **Orders** can contain **multiple items** with quantity and price tracking
- **Payment** transactions include **status tracking**
- Order and **shipping status** must be monitored

03

PRODUCT MANAGEMENT

- **Products** organized into **categories** with stock tracking
- Support for **promotions** with start/end dates and discounts
- Product details include images, descriptions, and inventory

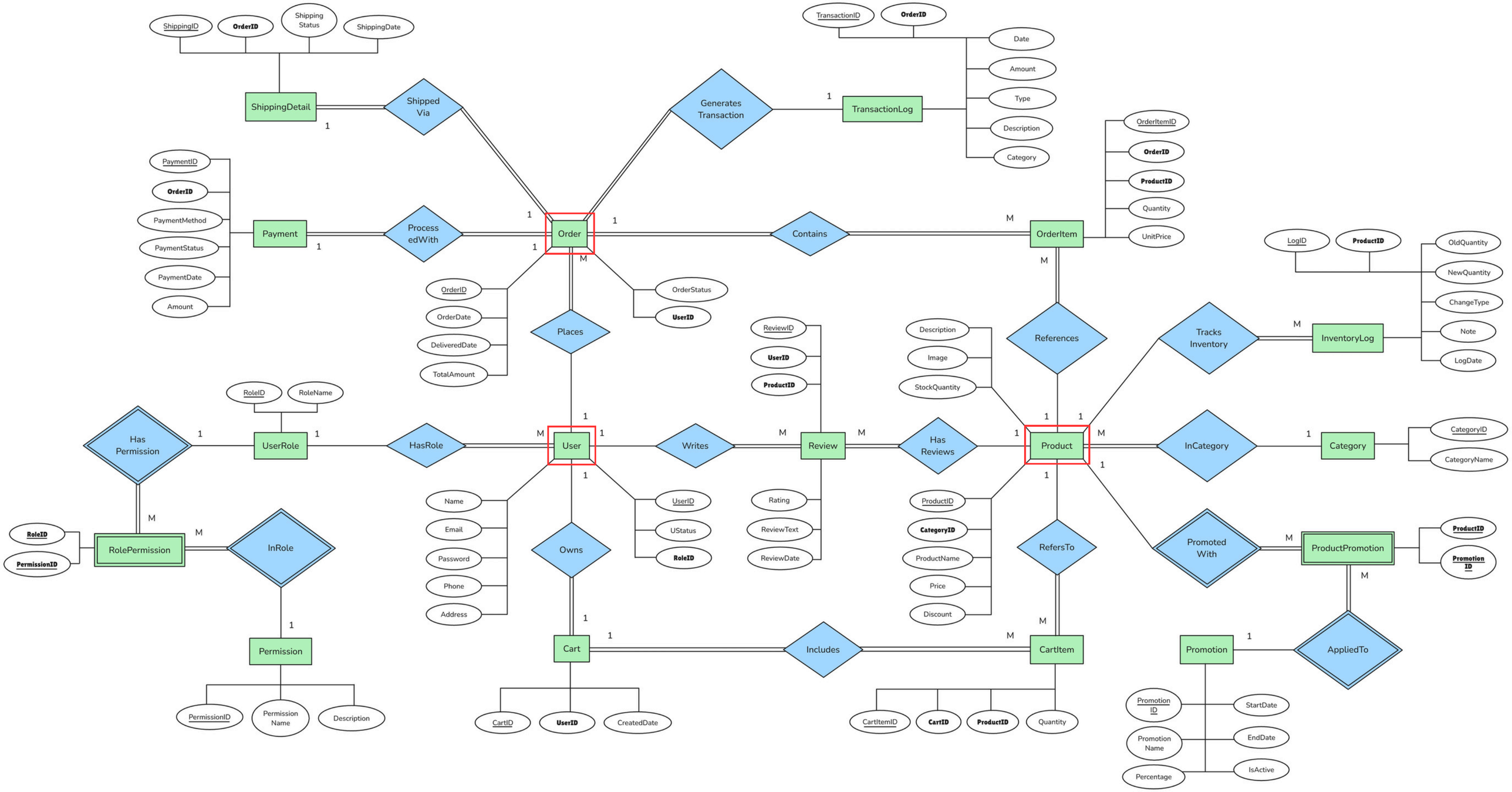
04

ADDITIONAL FEATURES

- **Shopping cart** functionality
- Product **review system** with ratings and comments
- Shipping addresses managed through user profiles
- **Reporting** capabilities for administrators
- Financial **transaction logging** and **inventory change tracking**

DATABASE DESIGN

ER Diagram Highlights



17 Entities, 18 Relationships. **Three main components** following requirements: **user** management, **order** tracking, **product** management

NORMALIZATION

1NF - No multivalued attributes
2NF - No partial functional dependencies
3NF - No transitive functional dependencies

01

1ST NORMAL FORM

From our design, **no multivalued attributes exists**. All schemas already in 1NF

02

2ND NORMAL FORM

All schemas has only **one PK** except RolePermission & ProductPromotion, which **does not have non-key attributes**. Hence, all are in 2NF already.

02

3RD NORMAL FORM

There **exists not any non-key attributes that depend on another non-key attributes**. Therefore, there's no transitive FD, hence all are in 3NF.

2NF Normalization Example

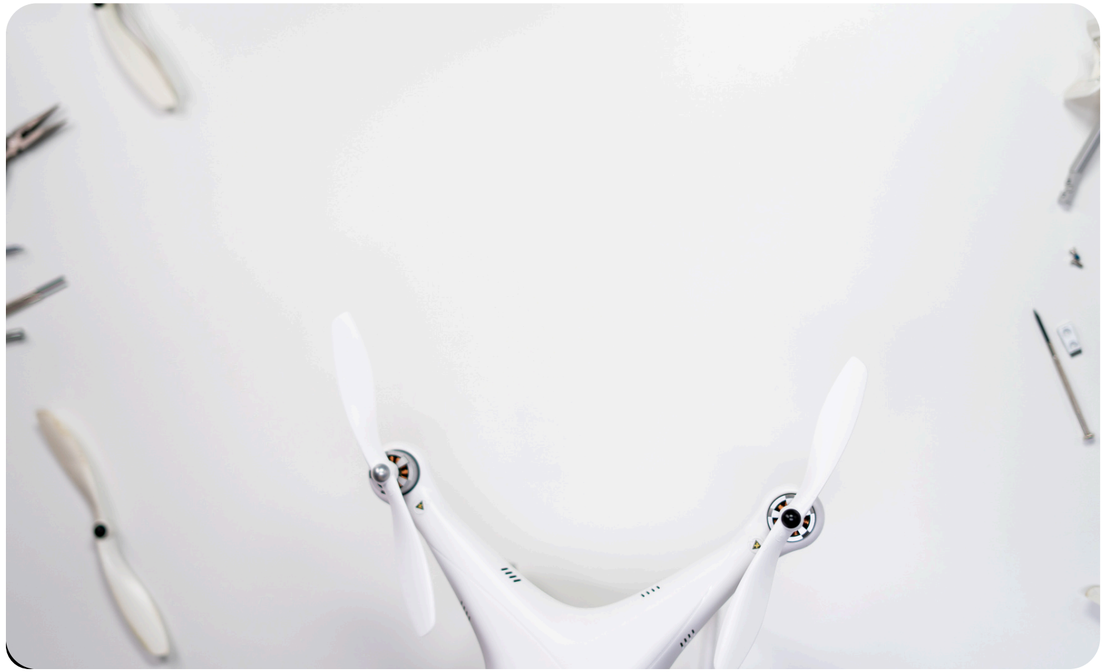
1. RolePermission(**RoleID FK**, **PermissionID FK**)
    - Partial functional dependency ไม่สามารถเกิดขึ้นได้ เนื่องจากไม่มี non-key attributes
2. ProductPromotion(**ProductID FK**, **PromotionID FK**)
    - Partial functional dependency ไม่สามารถเกิดขึ้นได้ เนื่องจากไม่มี non-key attributes

3NF Normalization Example

9. Cart(**CartID**, UserID FK, CreatedDate)
    - Functional Dependency: CartID -> {UserID, CreatedDate}
    - Transitive functional dependency ไม่สามารถเกิดขึ้นได้ เนื่องจากไม่มี non-key attribute ใดที่สามารถ uniquely identify non-key attribute อื่น (UserID ไม่สามารถ uniquely identify CreatedDate)

TECH STACK

Technology used to implement our project



01

FRONTEND

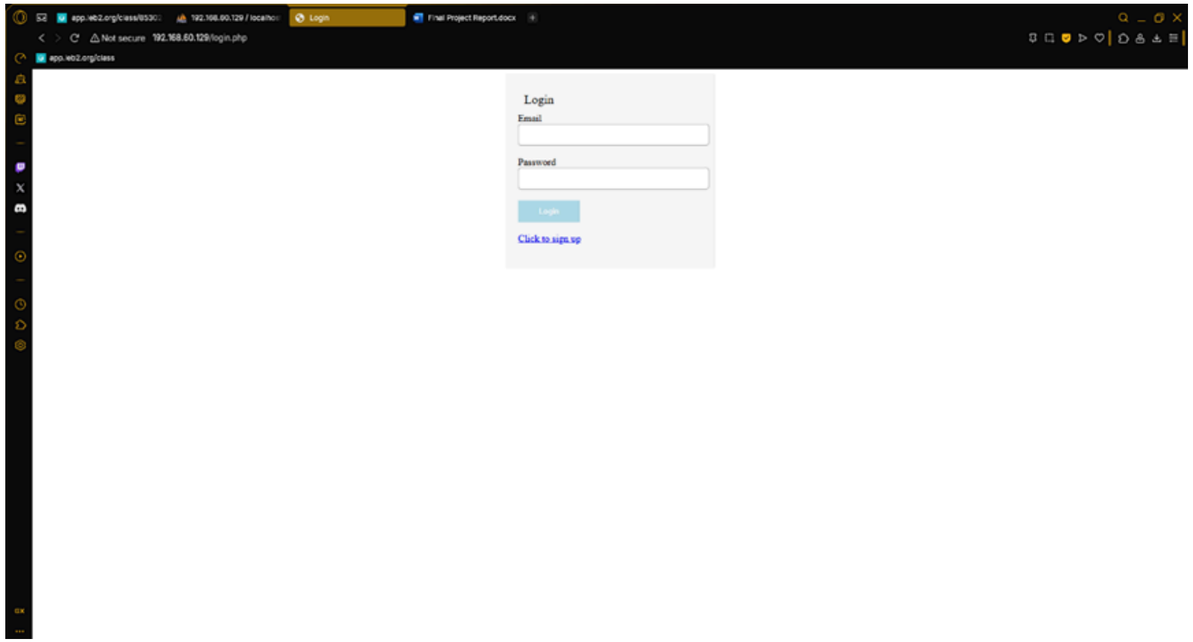
- HTML + CSS + Bootstrap
- Ease of use and simplicity, fitting for our project

02

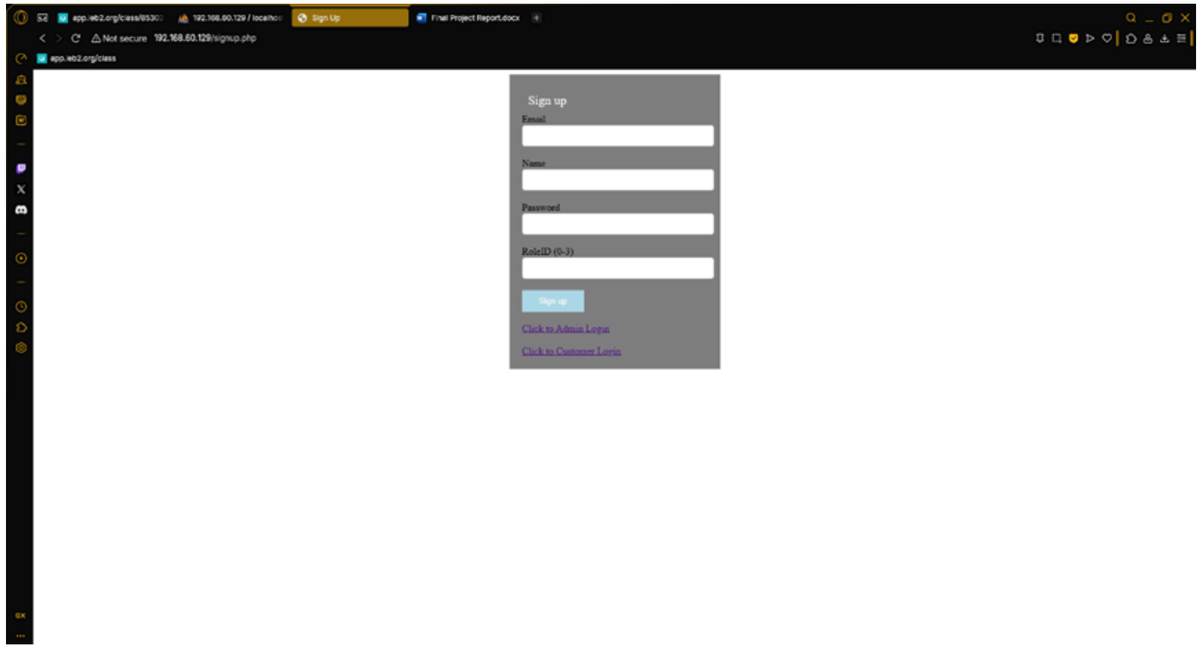
BACKEND

- PHP + MySQL on Ubuntu 24.04.1 LTS
- Implementation of our database design

UI OVERVIEW

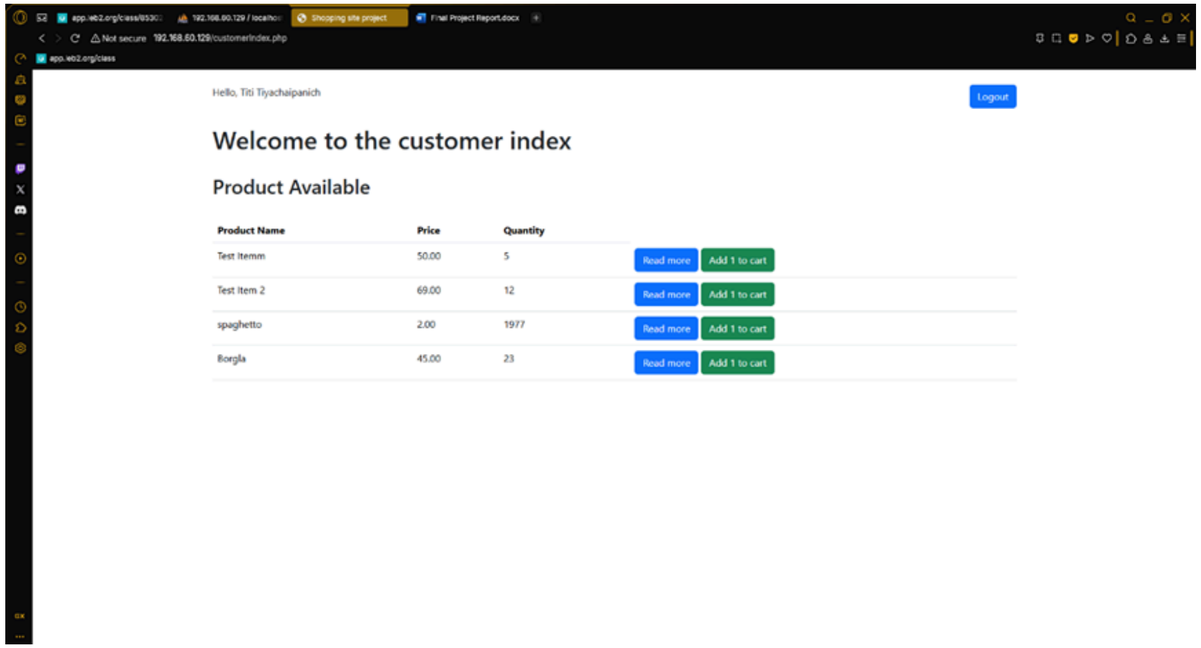


Login Page

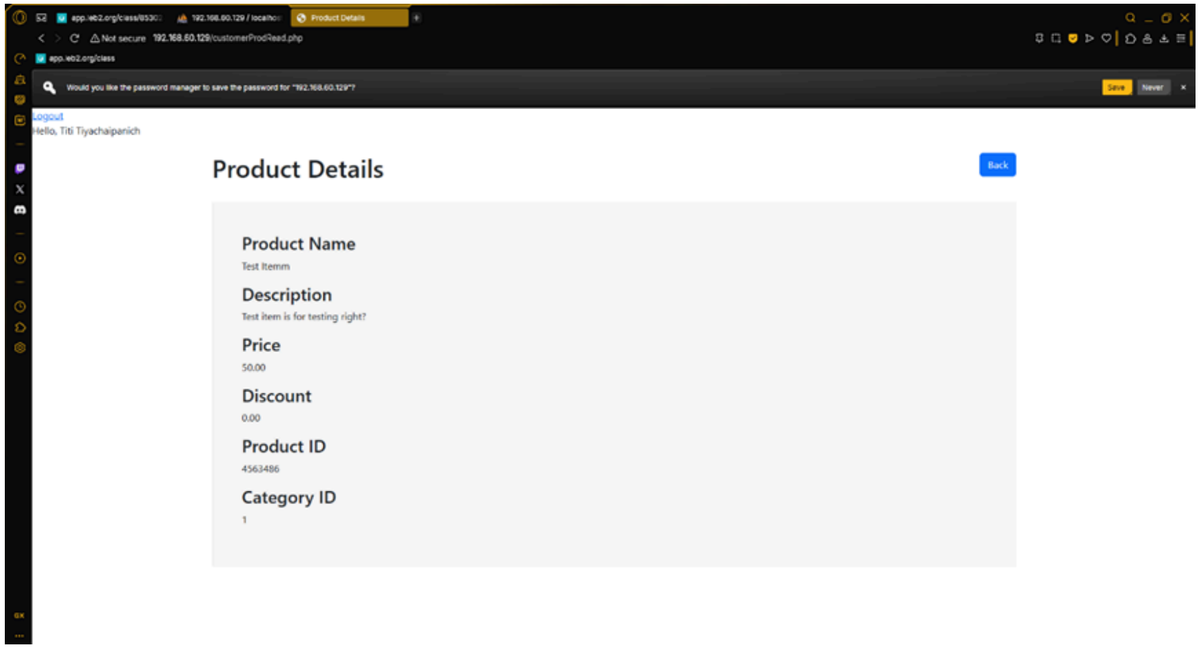


Register Page

Landing Page

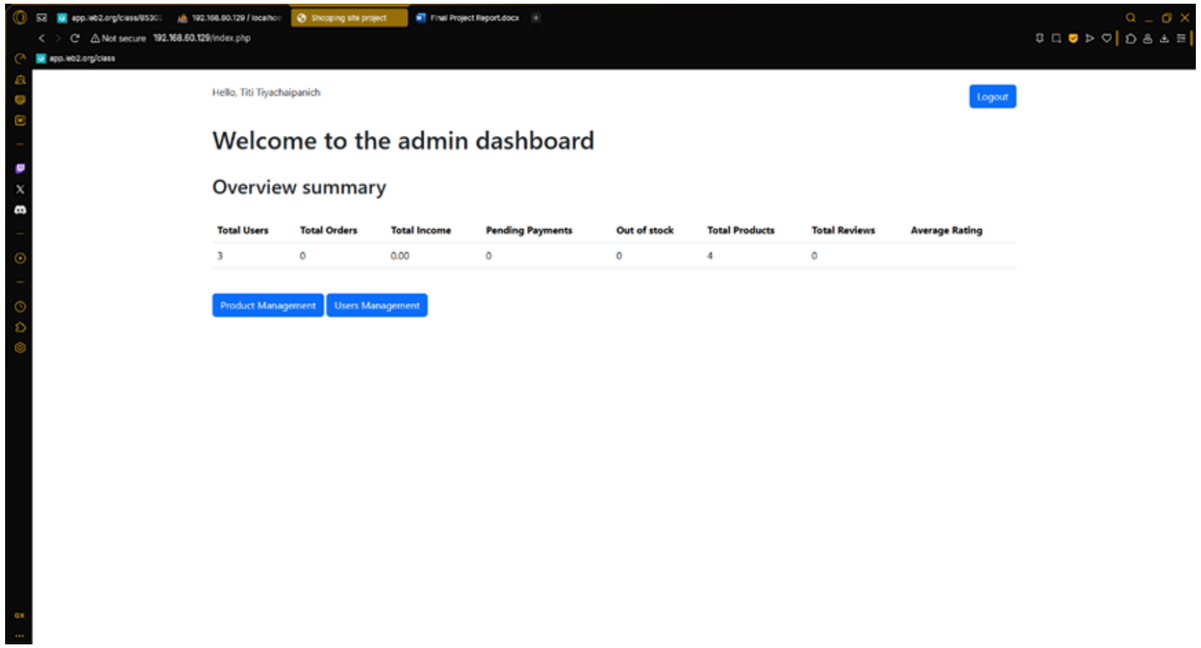


Choosing Product

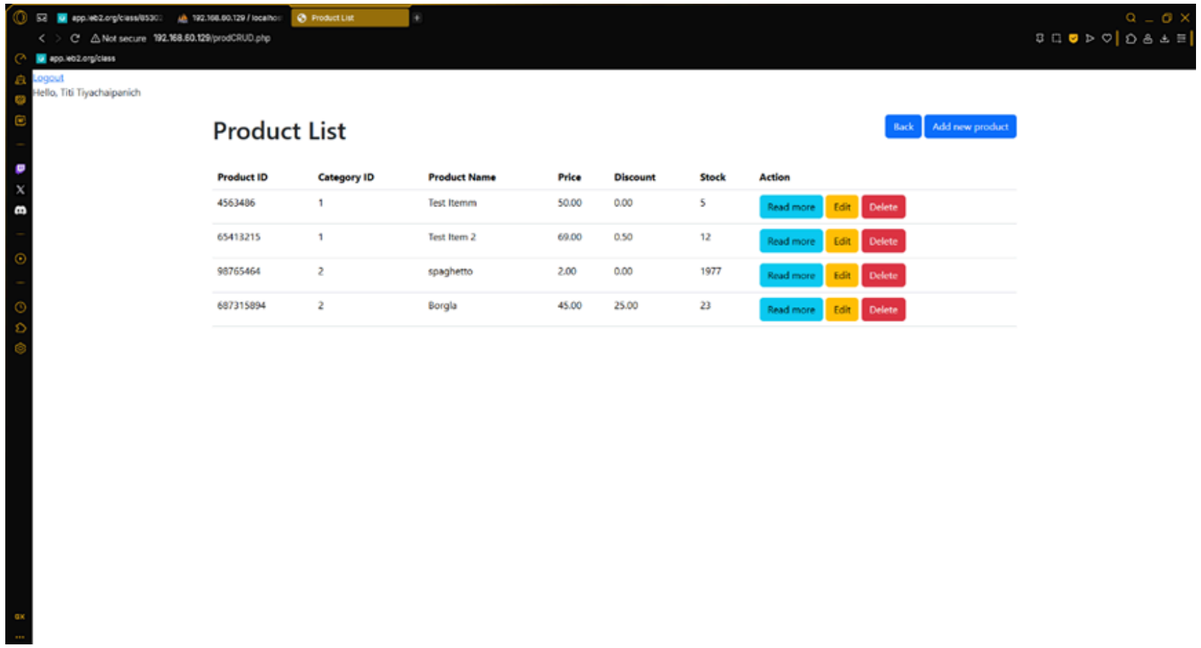


View Product Detail

Customer Page

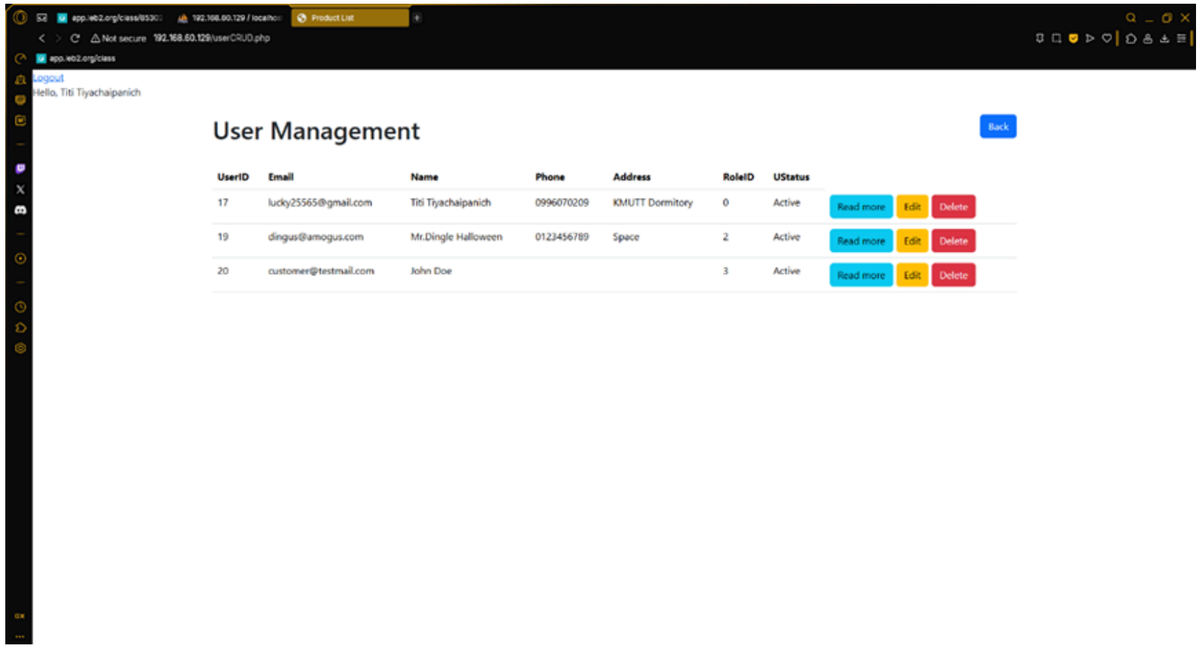


Dashboard Page

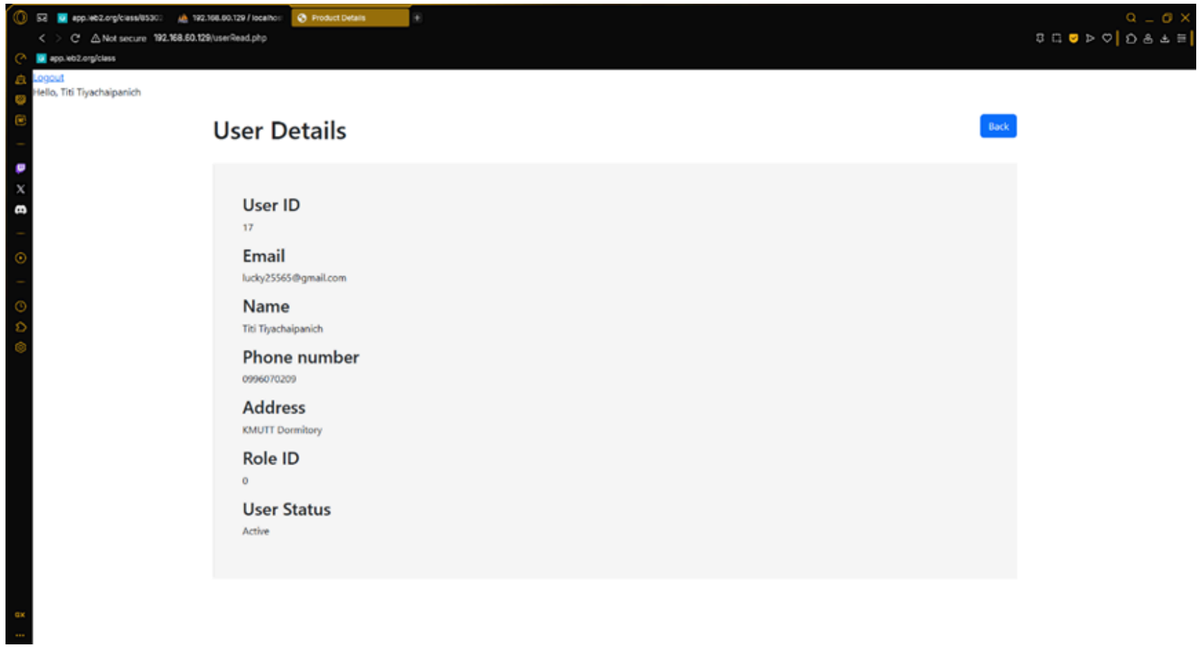


Product
Management Page

Admin Page

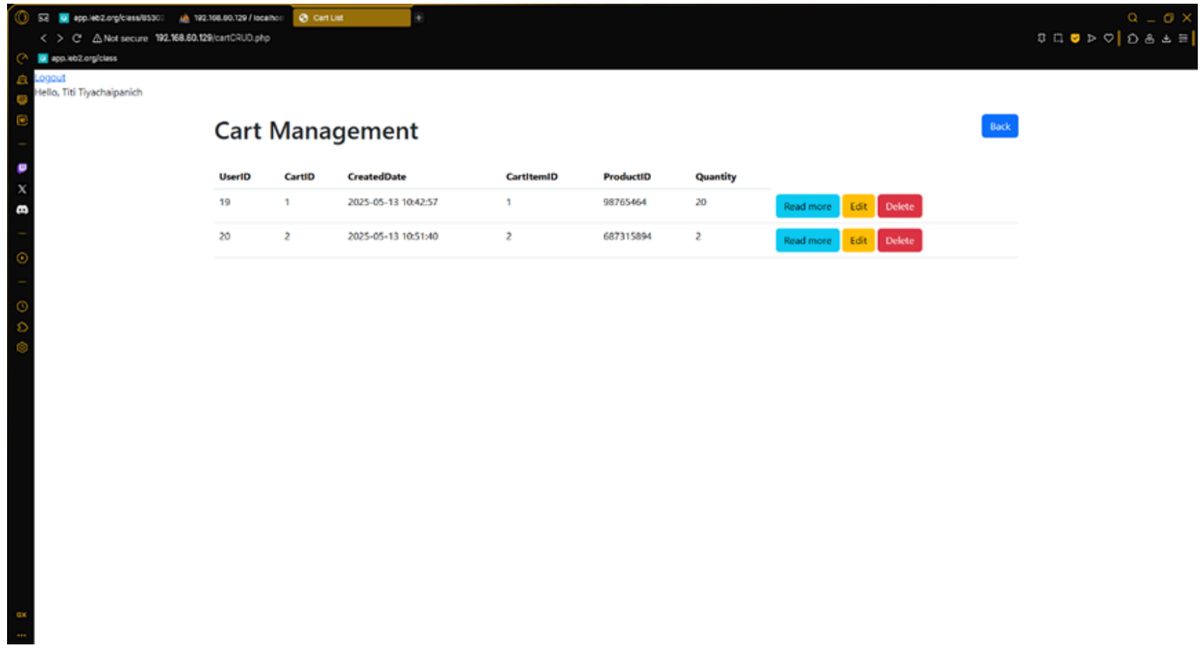


User
Management Page

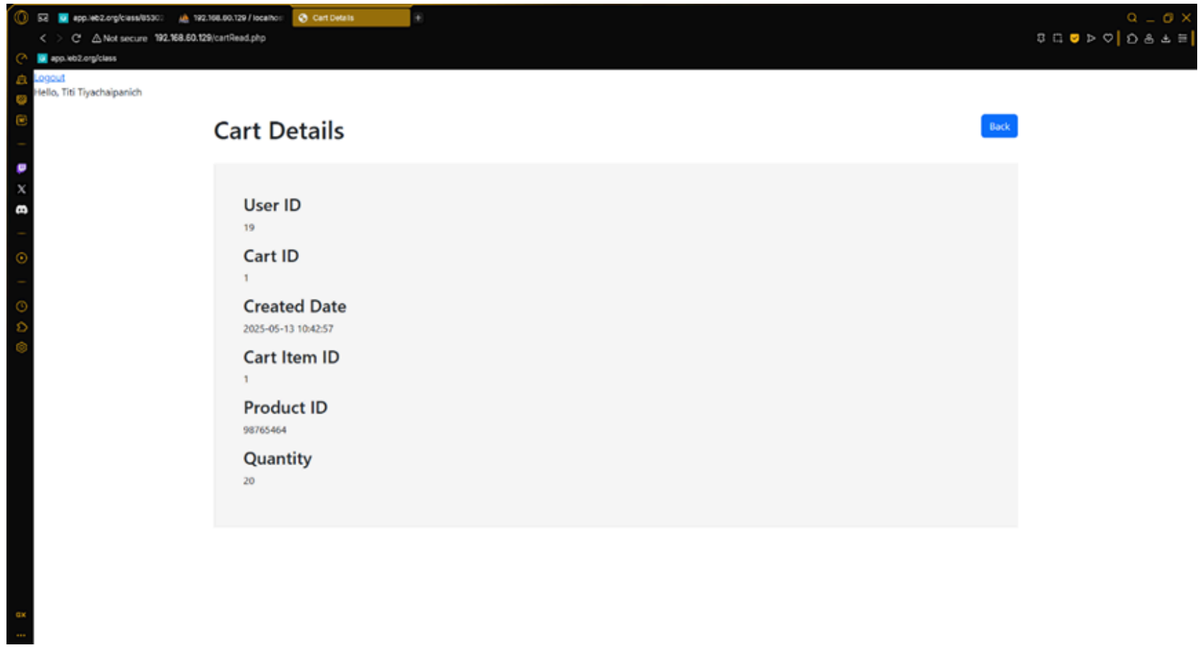


User Detail

Admin Page



Cart
Management Page



Cart Detail

Admin Page

QUERY DEMONSTRATION

```sql
--หาสินค้าด้วยการ search
SELECT
    ProductID, ProductName, Price, Image
FROM Product
WHERE ProductName LIKE '%?%';
```

สำหรับการค้นหาสินค้า
ด้วย Keyword

←T→	ProductID	ProductName	Price	Image
Edit Copy Delete	4	Lipstick Red	9.99	lipstick.jpg

```sql
SELECT
    ci.CartItemID,
    p.ProductName,
    ci.Quantity,
    p.Price,
    COALESCE(pr.Percentage, 0) AS DiscountPercent,
    ROUND(p.Price * (1 - COALESCE(pr.Percentage, 0)/100), 2) AS DiscountedPrice,
    ROUND(ci.Quantity * p.Price * (1 - COALESCE(pr.Percentage, 0)/100), 2) AS Subtotal
FROM CartItem ci
JOIN Cart c ON ci.CartID = c.CartID
JOIN Product p ON ci.ProductID = p.ProductID
LEFT JOIN ProductPromotion pp ON p.ProductID = pp.ProductID
LEFT JOIN Promotion pr ON pp.PromotionID = pr.PromotionID AND pr.IsActive = TRUE
WHERE c.UserID = 5;
```

การหา Total ใน Cart

CartItemID	ProductName	Quantity	Price	DiscountPercent	DiscountedPrice	Subtotal
3	Microwave 900W	1	120.00	5.00	114.00	114.00

```
--decrease when add to cart
DELIMITER //

CREATE TRIGGER trg_reduce_stock_on_add
AFTER INSERT ON CartItem
FOR EACH ROW
BEGIN
    UPDATE Product
    SET StockQuantity = StockQuantity - NEW.Quantity
    WHERE ProductID = NEW.ProductID;
END;
//

DELIMITER ;

--increase when delete from cart
DELIMITER //

CREATE TRIGGER trg_restore_stock_on_remove
AFTER DELETE ON CartItem
FOR EACH ROW
BEGIN
    UPDATE Product
    SET StockQuantity = StockQuantity + OLD.Quantity
    WHERE ProductID = OLD.ProductID;
END;
//

DELIMITER ;
```

ProductID	CategoryID	ProductName	Price	Discount	Description	Image	StockQuantity
1	1	Smartphone X	799.99	10.00	Latest model smartphone	smartphone.jpg	100

ProductID	CategoryID	ProductName	Price	Discount	Description	Image	StockQuantity
1	1	Smartphone X	799.99	10.00	Latest model smartphone	smartphone.jpg	125

s:uu Cart

OPTIMIZATION IDEA

```sql
-- ประวัติการซื้อของลูกค้า
SELECT * FROM `Order` WHERE UserID = ? ORDER BY OrderDate DESC;

-- เราสามารถใช้
CREATE INDEX idx_order_user_date ON `Order`(UserID, OrderDate);
```

```sql
-- รายการสินค้าในตะกร้า
SELECT * FROM CartItem WHERE CartID = ?;

-- เราสามารถใช้
CREATE INDEX idx_cartitem_cartid ON CartItem(CartID);
```

```sql
-- ข้อมูล Promotion
SELECT p.ProductName, pr.Percentage
FROM Product p
JOIN ProductPromotion pp ON p.ProductID = pp.ProductID
JOIN Promotion pr ON pp.PromotionID = pr.PromotionID
WHERE pr.IsActive = TRUE;

-- เราสามารถใช้
CREATE INDEX idx_productpromo_productid ON ProductPromotion(ProductID);
CREATE INDEX idx_productpromo_promotionid ON ProductPromotion(PromotionID);CREATE INDEX idx_cartitem_cartid ON CartItem(CartID);
```

CHALLENGE AND LESSON LEARNED

CHALLENGE

Lack of experienced member on web development

The cart - stock relation

LESSON

We should divide time more efficiently.

Q&A